

Lab 06: Build a Secure Google Cloud Network - Challenge Lab

Challenge Lab | Build a Secure Google Cloud Network | GSP322

Overview

Applied all skills from the Build a Secure Google Cloud Network badge to secure a real-world website (juice-shop) for a small business owner. Removed overly permissive firewall rules and implemented a secure bastion host architecture using IAP.

Scenario

Jeff owns a juice shop website running on Google Cloud. The previous engineer left the setup insecure with overly permissive firewall rules. The goal was to lock down access so that only authorized traffic reaches each server.

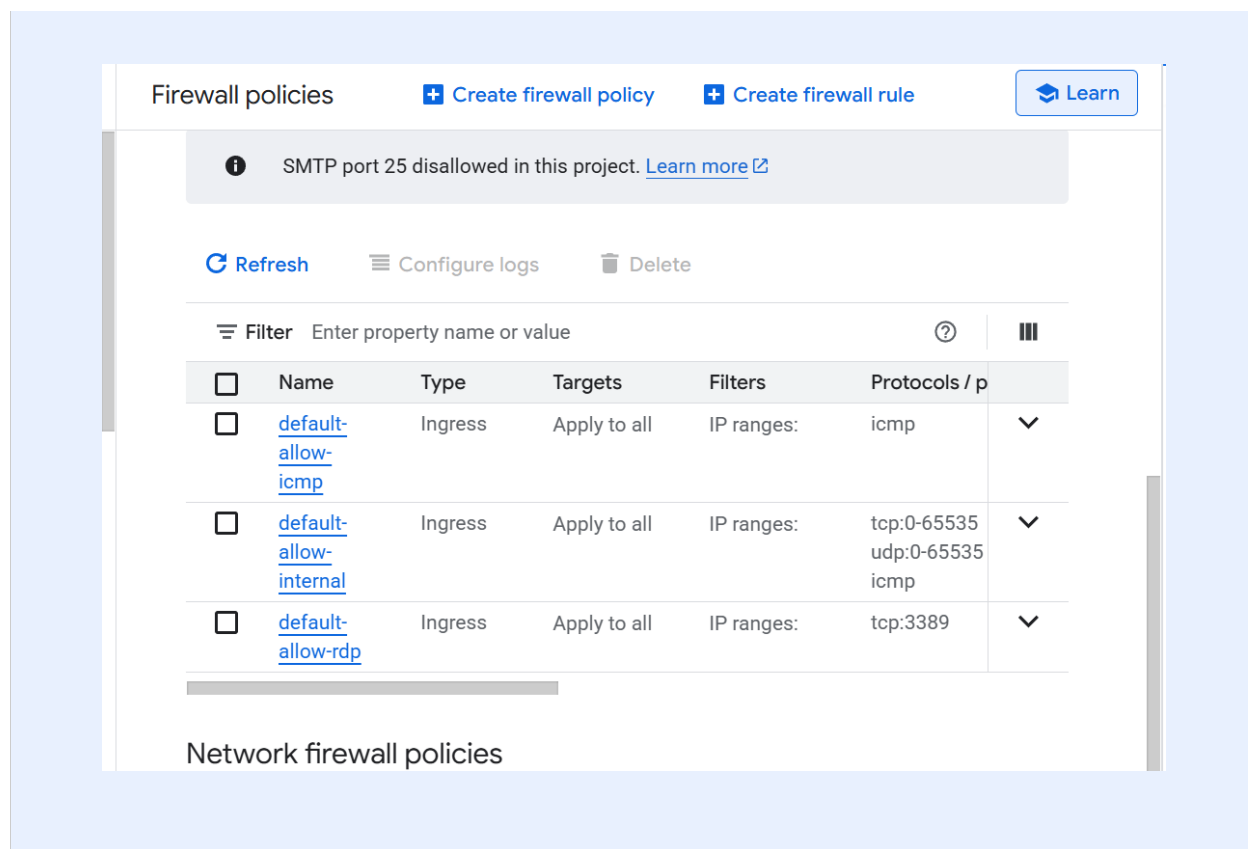
Security Requirements

Requirement	Solution
Bastion has no public IP	Removed external IP from bastion
SSH to bastion via IAP only	Firewall rule: IAP range 35.235.240.0/20 TCP 22
SSH to juice-shop via bastion only	Firewall rule: acme-mgmt-subnet range TCP 22
HTTP open to world for juice-shop	Firewall rule: 0.0.0.0/0 TCP 80 on http tag

Tasks Completed

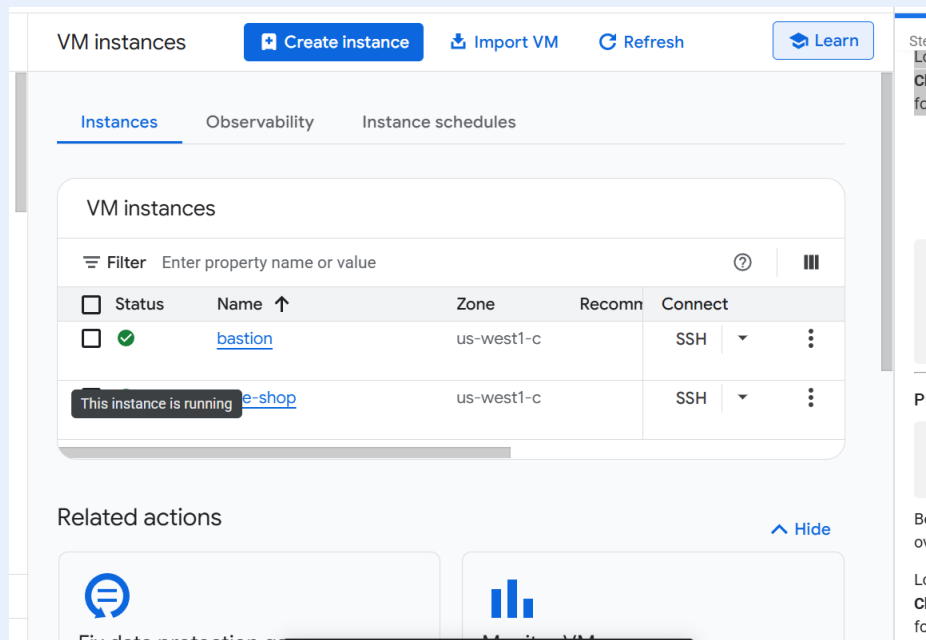
Task 1 - Remove Overly Permissive Rules

Identified and deleted firewall rules that allowed unrestricted access. These rules violated the principle of least privilege.



Task 2 - Start the Bastion Host

Located the stopped bastion host in Compute Engine and started it.



Task 3 - Create IAP Firewall Rule for Bastion SSH

Created a firewall rule allowing SSH only from the IAP IP range to the bastion host.

- Source range: 35.235.240.0/20 (Google IAP range)
- Protocol: TCP port 22
- Target tag: [SSH IAP network tag from lab]

```
gcloud compute firewall-rules create allow-ssh-iap-bastion \
  --network=[NETWORK] \
  --allow=tcp:22 \
  --source-ranges=35.235.240.0/20 \
  --target-tags=[SSH-IAP-TAG]
```

Firewall policies

Create firewall policy

Create firewall rule

Learn

Refresh

Configure logs

Delete

Filter

Enter property name or value

?

|||

☐	Name	Type	Targets	Filters	
☐	allow-ssh-iap	Ingress	grant-ssh-iap-ingress-ql-725	IP ranges:	▼
☐	default-allow-icmp	Ingress	Apply to all	IP ranges:	▼
☐	default-allow-internal	Ingress	Apply to all	IP ranges:	▼
☐	default-allow-rdp	Ingress	Apply to all	IP ranges:	▼

Network firewall policies

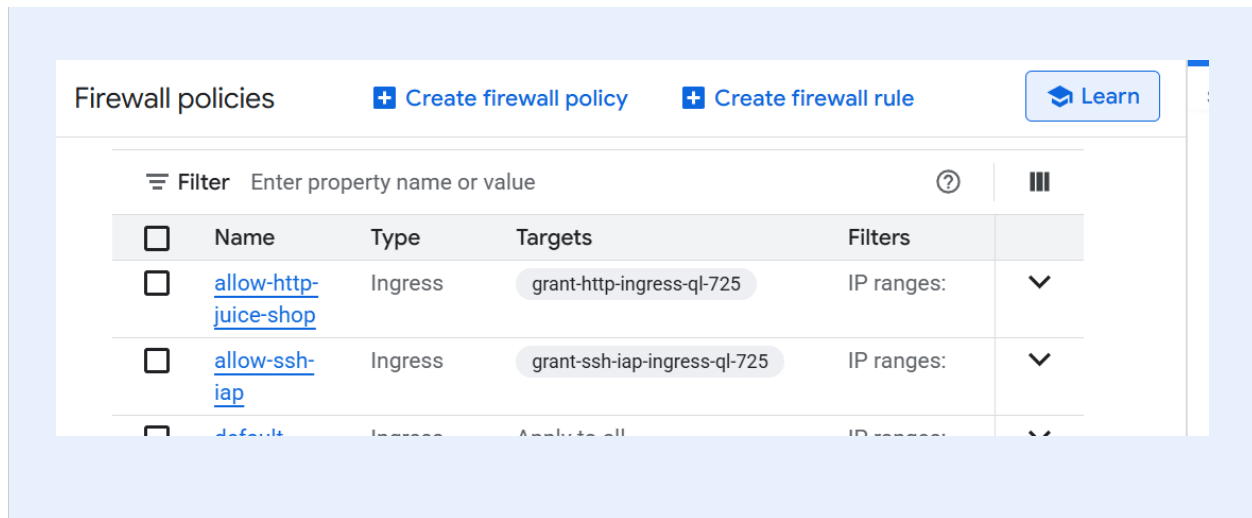
Firewall policies let you group several firewall rules so that you can update them all

Task 4 - Create HTTP Firewall Rule for juice-shop

Created a firewall rule allowing HTTP traffic from anywhere to the juice-shop server.

- Source range: 0.0.0.0/0
- Protocol: TCP port 80
- Target tag: [HTTP network tag from lab]

```
gcloud compute firewall-rules create allow-http-juice-shop \
  --network=[NETWORK] \
  --allow=tcp:80 \
  --source-ranges=0.0.0.0/0 \
  --target-tags=[HTTP-TAG]
```



Task 5 - Create Internal SSH Rule for juice-shop

Created a firewall rule allowing SSH from the management subnet to juice-shop so the bastion can connect to it internally.

- Source range: acme-mgmt-subnet CIDR range
- Protocol: TCP port 22
- Target tag: [SSH internal network tag from lab]

```
gcloud compute firewall-rules create allow-ssh-internal-juice-shop \
  --network=[NETWORK] \
  --allow=tcp:22 \
  --source-ranges=[MGMT-SUBNET-CIDR] \
  --target-tags=[SSH-INTERNAL-TAG]
```

Firewall policies [+ Create firewall policy](#) [+ Create firewall rule](#) [Learn](#)

i SMTP port 25 disallowed in this project. [Learn more](#)

[Refresh](#) [Configure logs](#) [Delete](#)

Filter Enter property name or value ? |||

☐	Name	Type	Targets	Filters	
☐	allow-http-juice-shop	Ingress	grant-http-ingress-ql-725	IP ranges:	▼
☐	allow-ssh-iap	Ingress	grant-ssh-iap-ingress-ql-725	IP ranges:	▼
☐	allow-ssh-internal	Ingress	grant-ssh-internal-ingress-ql-725	IP ranges:	▼
☐	default-allow-	Ingress	Apply to all	IP ranges:	▼

Task 6 - SSH to Bastion and juice-shop

Connected to the bastion via IAP tunnel then SSH'd from the bastion to juice-shop internally.

```
gcloud compute ssh bastion --tunnel-through-iap --zone=[ZONE]
ssh juice-shop
```

```
Linux bastion 5.10.0-43-cloud-amd64 #1 SMP Debian 5.10.251-5 (2026-05-15) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-00-997c36971fff'.
student-00-997c36971fff@bastion:~$ ssh 192.168.11.2
The authenticity of host '192.168.11.2 (192.168.11.2)' can't be established.
ECDSA key fingerprint is SHA256:umdQZ7ptGHBTwSWzBL+A8HGkk4dK1TyLbOf5jK6W6Ns.
Are you sure you want to continue connecting (yes/no/[fingerprint])? ^C
student-00-997c36971fff@bastion:~$ yes
y
y
```

```

This tool needs to create the directory [/home/student-00-997c36971fff/.ssh] before being able to generate SSH
keys.

Do you want to continue (Y/n)? gcloud compute ssh juice-shop --zone=us-west1-c --192.168.11.2
Please enter 'y' or 'n': Y

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/student-00-997c36971fff/.ssh/google_compute_engine
Your public key has been saved in /home/student-00-997c36971fff/.ssh/google_compute_engine.pub
The key fingerprint is:
SHA256:N2unFPyrkepB/H6GoCAGm4bYUj/FzZLTiuVWArJEgK4 student-00-997c36971fff@bastion
The key's randomart image is:
+---[RSA 3072]-----+
|  ..+o               |
|  .  .               |
|  .  . *             |
|  .  . O.B.          |
|oB . = BS =         |
|E.+ = +..o *        |
|.o . + ...B.o       |
|  .  . *+.o         |
|  .o ++             |
+-----[SHA256]-----+
Linux juice-shop 5.10.0-43-cloud-amd64 #1 SMP Debian 5.10.251-5 (2026-05-15) x86_64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
sa_115103234089141374436@juice-shop:~$

```

What I Learned

- Overly permissive firewall rules are a major security risk in cloud environments
- Bastion hosts act as a secure jump box for accessing private VMs
- IAP tunnels eliminate the need for a public IP on the bastion host
- Network tags allow precise targeting of firewall rules to specific VMs
- The principle of least privilege means only allowing exactly what is needed
- Zero trust means verifying identity before allowing any network access

Challenges & How I Solved Them

- **Overly permissive firewall rules** — The default open-access rule was allowing unrestricted traffic. I removed it and replaced it with specific rules scoped to the correct source ranges and target tags.
- **IAP SSH connection failing (Error 4003)** — The browser SSH button was trying to connect directly instead of through IAP. I solved this by using the "**Open in browser window with IAP tunnel**" option from the SSH dropdown menu on the bastion VM.
- **Network tags not applied** — The firewall rules were correctly configured but weren't taking effect because the network tags weren't saved on the VMs. I fixed this by editing each VM and ensuring the tags were properly committed.
- **SShing from bastion to juice-shop** — A plain ssh <internal-ip> command wasn't reliable from inside the bastion. Using gcloud compute ssh juice-shop --zone=us-west1-c --internal-ip forced the connection through the internal network correctly.

Reflection

This challenge lab brought together VPC networking, IAP, firewall rules, and the bastion host pattern into one real-world security scenario. Completing all tasks proves practical knowledge of securing Google Cloud network infrastructure.

Resources

GCP VPC Firewall Docs: <https://cloud.google.com/vpc/docs/firewalls>

GCP IAP Docs: <https://cloud.google.com/iap/docs>

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